

Frederick Robinson

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Employment

- 2024–2026 **Principal Applied Scientist (later Member of Technical Staff)**, *Amazon SF AI Lab*
- Founding member of the reinforcement learning group, standing up RL training systems.
 - Built [Turnstile \(Amazon Science\)](#), an open-source Rust proxy capturing token-level rollout data from arbitrary black-box agent harnesses, with prefix-aware sequence collapsing and verified loss masks.
 - Reproduced verifier-free RL (VeriFree, RLPR) and located its failure boundary: rewards from reference-answer likelihood degrade once answers lengthen or admit paraphrase.
 - Drove data research for RLVR post-training: dataset ablations, offline difficulty calibration (16 samples vs. SOTA), and targeted acquisition against eval failure clusters.
- 2024 **Member of Technical Staff**, *Adept*
- Improved reliability of computer-use agents by developing evaluation frameworks, data collection pipelines, and SFT experiments.
 - Contributed foundational work that supported the team's transition into Amazon.
- 2022–2023 **Staff Machine Learning Engineer**, *Stripe, Inc. – Supportability Intelligence*
- Trained a transformer classifier over merchant website text, and led an LLM-based enforcement initiative and cross-functional team delivering > 20% improvement in Terms of Service violation detection.
- 2017–2022 **Senior Applied Scientist**, *Twitch (Amazon AWS Interactive Video Service)*
- Tech lead for Video Intelligence. Built production ML at scale for data center selection (> 1% global buffering reduction) and real-time viewbot detection (> 75%).
- 2014–2017 **Software Engineer**, *Google, Inc. – Display Ads, Remarketing Quality*
- Built ML and real-time bidding systems unlocking tens of millions in annual revenue.

Skills

- ML Large Language Models, Reinforcement Learning (CoT, RLVR), Supervised Fine-Tuning, Evaluation Frameworks, Distributed Training Systems, Data Collection Pipelines
- Tools PyTorch, slime, SGLang, Python, Golang

Education

- 2014 **Master of Arts in Mathematics**, *University of California Los Angeles*
- 2012 **Bachelor of Arts in Mathematics with Honors**, *Northwestern University*